

Agent Handbook — 01 Revenue-Cycle & Denial

The flagship land agent. Drafts denial work product and root-causes denial codes for a provider's revenue-cycle team — and **cannot submit a claim**. Maturity: Demonstrated + Deployable-by-design; production-readiness (CSV/CSA, IdP, live connectors, pen test, HITRUST) is the engagement; AWS BAA precedes any PHI.

Workflow + human gate

Ingest a denied/remittance record → classify the denial (e.g. CO-197 no/invalid prior auth, CO-50 not medically necessary, CO-16 missing/incorrect info) → pull supporting documentation through governed connectors → ground every cited code/amount/policy → **draft** the appeal packet and a root-cause summary → **suspend at the Denials-Specialist gate** for review and submission. **Named human gate:** Denials Specialist.

Systems / connectors (with standards)

- **EHR / FHIR** — Epic / Oracle Health, normalized via Amazon HealthLake (FHIR R4): chart, encounter, coverage.
- **Clearinghouse** — Change Healthcare / Availity / Waystar: **X12 837** (claim), **835** (remittance/denial), **277** (claim status).
- **Knowledge base** — payer policy / denial-reason corpus for grounding.

Workforce roles to map in the IdP

DENIALS_SPECIALIST (the gate), DENIALS_MANAGER (escalation/oversight), BILLER (read/assist). Map these to custom:hpp_role in Cognito at engagement time.

Withheld authority (what it never does)

Never submits a claim — clearinghouse.submit_claim is **withheld from the agent**. It drafts the appeal; the Denials Specialist submits. No write to the system of record without the gate.

Deploy (fixture mode, no PHI)

```
scripts/build_lambdas.sh
scripts/deploy.sh 01-revenue-cycle-denial dev portable native
s3://my-cfn-staging/hpp 10.30.0.0/16
```

- Params: infra/cloudformation/params/01-revenue-cycle-denial.json · Terraform: terraform apply -var-file=envs/01-revenue-cycle-denial.tfvars
- VPC CIDR 10.30.0.0/16. Gateway portable (use agentcore only where available); runtime native (Step Functions) or container (Fargate).

Human-gate smoke test

1. Sign in as a seeded DENIALS_SPECIALIST.

2. Drive a gated write (draft an appeal packet) and confirm the run **suspends** at the Denials-Specialist gate; a HITL item appears.
3. Approve as the specialist → action proceeds, approval written to the **append-only PHI-masked audit**.
4. Reject → action does **not** proceed.
5. Attempt clearinghouse.submit_claim → gateway **denies by default** (withheld).

Key regs

HIPAA Privacy/Security, HITECH; payer appeal rules; False Claims Act / OIG exposure on coding accuracy; CMS-0057-F upstream (denials driven by CO-197 prior-auth gaps).

Go-live checklist

- ☐ AWS BAA + SI BAA executed; no PHI before both.
- ☐ IdP federated; DENIALS_SPECIALIST/MANAGER/BILLER mapped to custom:hpp_role.
- ☐ Live connectors validated (Epic/Oracle Health; Change Healthcare/Availity/Waystar; X12 837/835/277).
- ☐ Guardrail thresholds + grounding corpus tuned to payer policy.
- ☐ CSV/CSA validation signed off; HITL queue SLA ratified.
- ☐ Pen test + HITRUST/SOC 2 evidence assembled; observability + runbooks adopted.